

Group-aware validation sharply reduces the apparent accuracy of extraction-category classification in public cannabis laboratory data

Arthur Gonçalves Breguez

Independent researcher

ORCID: 0009-0005-8551-731X

Correspondence: arthurbreguez@gmail.com

Preprint. Code and evidence logs: <https://github.com/ArtBreguez/cannabis-extraction-ds>

Abstract

Published models that classify cannabis products from cannabinoid and terpene profiles report near-perfect accuracy: above 95% for resin type, 100% for chemovar class. These figures come from random cross-validation or Kennard-Stone partitioning, which hold out no plant, producer or laboratory.

We quantify the cost on a task whose group structure is explicit. On 33,229 public laboratory-result rows for cannabis concentrates that name a producer, nearly all from Nevada, each with a source-declared extraction category (non-solvent against solvent-based), a gradient-boosted classifier scores 78.5% balanced accuracy and a ROC AUC of 0.89 under random five-fold cross-validation. Holding out the producer leaves 54.0% and 0.62. A signal survives: the pooled held-out AUC is 0.61, with a producer-level bootstrap interval of 0.535 to 0.698. But the reported model's pooled balanced accuracy does not separate from chance (51.6%, interval 47.3 to 56.6); it keeps about 30% of its above-chance AUC, and an unweighted logistic regression 41 to 53%.

A designed experiment shows the same direction: with the random split averaged over 200 seeds, 162 HPLC-assayed samples fall from 54.3% to 38.9% under leave-one-variety-out with gradient boosting and from 51.8% to 46.3% with logistic regression, against 33.3% chance. In the market data the same profiles identify the producer at 67.1% against a 14.4% baseline and the laboratory at 92.3% against 40.3%.

We conclude that the declared extraction category is only weakly recoverable from these analyte panels for an unseen producer, and that a random split overstates it by 8 to 25 points of balanced accuracy depending on the model. We did not rerun the published studies.

Keywords: cannabis, chemometrics, data leakage, group cross-validation, extraction category, laboratory testing data, model validation

1. Introduction

Cannabis chemical profiles are now routinely fed to supervised classifiers. Reported performance is consistently high. A support vector machine separates Dutch-type from Moroccan-type cannabis resin at 95.2% accuracy from two cannabinoid concentrations [1]. Partial least squares discriminant analysis as-

signs chemovar class from near-infrared spectra of whole inflorescences with "100% prediction accuracy", sensitivity and specificity both 1.00 [2]. A combined cannabinoid-and-terpene PLS-DA model reaches 0% prediction misclassification across chemovars [3].

These numbers invite a question the papers do not ask: performance on *what* unseen unit? A shuffled fold defines "unseen" as a row absent from the training partition. Deployment defines it more strictly, as a plant, a producer, or a laboratory the model has never encountered. When several rows descend from the same entity, a random split lets the model recognise the entity instead of learning the property of interest, and the reported score measures memorisation.

This failure mode is established in the general methodological literature [4, 5, 6, 7]; a related form, selection bias from choosing features outside the cross-validation loop, was quantified in genomics more than two decades ago [8]. It has been described specifically for chemometrics, where Király and Tóth show that the Kennard-Stone selection algorithm and the inappropriate handling of repeated measurements both introduce train/test leakage [9]. The cannabis applications cited above are exposed to it in different ways. Reference [2] grows 88 genotypes cloned in triplicate to obtain 264 plants, then partitions with Kennard-Stone [10] with no statement that the three clones of a genotype were kept in one partition, on a 231/33 class split. Reference [1] evaluates plain accuracy over five random 80/20 train/test iterations on a 149/40 class split, where always answering the majority class already scores 78.8%; its two classes also come from different countries and collection routes, retail purchases in the Netherlands against seizures in north-east Italy, so class is confounded with source. Reference [3] reports a 4% cross-validation error beside its 0% prediction error and its abstract does not say how its prediction set was drawn.

The bridge between the two literatures has not been built. We are not aware of a published cannabis study that holds out the producer, the genotype or the laboratory and reports what happens to the score.

We build that bridge on a concrete task. The question is whether a concentrate's cannabinoid and terpene profile carries enough information to recover how it was made, separating concentrates declared *non-solvent based*, which under the Nevada rule in force since November 2020 means mechanical separation (hashish, bubble hash and the like) or extraction with ethanol or CO₂ [11], from those declared *solvent based*, made with any other approved solvent. We keep the shorthand *solventless* and *hydrocarbon* for the two declared categories because the code and evidence logs use them; 2.1 reports what each category contains. The task is a good test case for three reasons. The label is declared at source rather than inferred. The group structure is explicit, since every row names a laboratory and most name a producer. And a substantial subset of producers make both classes, which permits a test where a held-out producer's label cannot be read off the training rows.

We report a largely negative result for the chemical question and a quantitative result for the methodological one.

2. Materials and methods

2.1 Data

Market data. The Cannlytics `cannabis_results` corpus [12] aggregates public cannabis laboratory results obtained through public records requests and from certificates of analysis published online, released under CC-BY-4.0. The retrieved file contains 808,407 rows, of which 139,714 are concentrates or extracts. None of the rows used here carries a certificate link, a batch number or a strain name, and their `product_type` values read like track-and-trace item categories (for example `Solvent Based Concentrate (Each)`), so we treat them as a state laboratory-result export and not as parsed certificates. The `producer` field is the licensee that the export records for the lot. We use it as the group to hold out, without being able to tell a cultivator from a processor or a packager.

Labels come from the `product_type` field, which in some markets declares the extraction family directly as `non-solvent based concentrate` or `solvent based concentrate`. This is a source-declared label, not an inference from a commercial product name. Rows whose `product_type` is only `concentrate` remain unlabelled and are never assigned to a class. The analyses run on the 37,344 labelled rows that survive the consistency check described below, which we call the usable rows. Most concentrate rows, 73.3% of them, carry no such declaration and are dropped, which is where the bulk of the reduction from 139,714 concentrate rows to 37,374 labelled rows happens (Section S1, Table S1).

A secondary labelling path applies regular expressions to `product_name` (`hash rosin`, `live rosin`, `rosin` for solventless; `bho`, `butane`, `hydrocarbon`, `live resin`, `shatter`, `badder`, `batter`, `wax` for hydrocarbon). The two paths can only be compared on the 11.5% of labelled rows whose names carry one of those words. Where both paths produce a label they agree on 4,259 of 4,289 rows, 99.3%. The 30 contradictory rows are excluded. The name path never adds a row: it only cross-checks the declared label, so every labelled row carries a `product_type` declaration.

The resulting labelled dataset is 37,374 rows, of which 30 carry contradictory labels and are excluded from every analysis, leaving **37,344 rows** that all figures below are computed on unless stated otherwise: 24,573 declared solvent-based (65.8%) and 12,771 declared non-solvent (34.2%). Of these, 33,229 (89.0%) name one of 96 producers, every row names one of 12 laboratories and none carries a strain name (Tables S2 and S3).

Every usable row that carries a `lab_state` is from Nevada: seven laboratories and 31,422 of the usable rows (84.1%, and 94.6% of the rows that name a producer). The field is blank for the remaining 5,922 rows, which come from five laboratories, four that record no producer and one that does. No row names another state, but the names of the four producer-less laboratories (one is `Cannalytics RI`) suggest they are not in Nevada, in which case the rule cited below does not govern their 4,115 rows. Tests are dated 2020-01-02 to 2024-03-29 on the 33,765 rows that carry a date; the producer-less rows are all dated from December 2022 on.

Rows are not samples. 419 of the rows that name a producer repeat another row exactly, 1,303 samples from the producer-less laboratories are stored as two adjacent rows each, 1,071 rows that name a producer report nothing but a `total_terpenes` of zero, and two of the 96 producer strings are spellings of one company (Section S2). We keep the rows as stored, since nothing marks which copy is the record, and 3.2

reports the headline pair without the duplicates, without the terpene-only rows and with the two spellings merged. The split pairs lie outside the 33,229 rows that name a producer, on which the headline comparison runs. Those rows are very unequally spread: the largest producer holds 13.2% of them, the five largest 42.3%, and the median producer 64 rows.

What the label means. The two categories are regulatory, not chemical. Nevada's rule separates the same pair, and in its versions of July 2022 and October 2023 defines it as "Extract of cannabis (nonsolvent) like hashish, bubble hash, infused dairy butter, mixtures of extracted products or oils or fats derived from natural sources, including concentrated cannabis extracted with ethanol or CO2" against "Extract of cannabis (solvent-based) made with any approved solvent, including concentrated cannabis extracted by means other than with ethanol or CO2" [11]. The text effective November 2020 carries the same pair of definitions, and the 2018 rule it replaced named CO2 only, so at least the first six months of the data, which precede the adoption of the current regulations in July 2020, fall under the older wording. Under the current definition the non-solvent category spans mechanical separation, ethanol and CO2, and the product names are consistent with that. Table 1 counts them over the rows that carry a name, which are the 33,229 rows with a producer; a name may hit several families.

Table 1. Product-name word families in each declared category, over the 33,229 named rows. Each family is a regular expression in `src/evaluation/label_composition.py`; besides the words listed, the hydrocarbon family matches `hydrocarbon` and `batter`, the mechanical one `solventless`, the ethanol one `etoh` and the distillate one `disty`.

word family in product_name	non-solvent	solvent-based
distillate	5,805 (51.5%)	8,258 (37.6%)
co2 / supercritical	225 (2.0%)	19 (0.1%)
ethanol	163 (1.4%)	214 (1.0%)
hydrocarbon words (bho, butane, live resin, shatter, badder, wax)	43 (0.4%)	3,805 (17.3%)
mechanical words (rosin, hash, kief, bubble, ice water)	559 (5.0%)	114 (0.5%)

Only 5.0% of the named non-solvent rows contain a mechanical word, while 51.5% are named `distillate`, a product that presupposes a crude extract and is not how mechanical concentrates are sold, 2.0% name CO2 and 1.4% name ethanol. The word families are crude. All 43 non-solvent rows that hit a hydrocarbon word also say rosin, where badder and wax are consistencies, and the 114 solvent-based rows that hit a mechanical word contain hash, bubble or kief inside cultivar names, none of them rosin. The table describes the two categories; it cannot count mislabelled rows.

Neither the wording of the rule nor practice was constant over the period, and they did not change together. Product names that say ethanol are almost all solvent-based through 2022 (201 of 211) and mostly non-solvent in 2023 (133 of 146; Table S4 in Section S3), so the meaning of the label drifts within the data.

The task this paper tests is therefore the regulator's category as the laboratories applied it over the four years, 2020 to 2023, that the rows naming a producer cover, mechanical, ethanol or CO2 extraction against the remaining solvents, and not rosin against BHO specifically; 3.2 reports the headline pair on two narrower subsets.

Controlled data. Zenodo record 13823859 [13] is a designed experiment rather than market data: six named varieties of non-psychoactive, CBD-type cannabis, each extracted by three laboratory methods (maceration, ultrasound, supercritical CO₂) under varied time, temperature and pressure, with cannabinoids quantified by HPLC. The record's description lists Soxhlet as the third method; the data file labels it `Ultrasonido`, which we follow. The design is fully balanced at 54 samples per method and 27 per variety, 162 total, with four features (CBD, CBG, CBN, THC) and no missing values. Each sample is a distinct (method, variety, time, temperature, pressure) cell, so there are no replicates to leak across a random split; the chemotype also limits transfer to THC-dominant concentrates. The process settings (time, temperature, pressure) are not features: pressure alone identifies supercritical CO₂, and adding the three settings lifts the random split to 88.3%.

2.2 Feature construction

The source file carries 34 analyte columns. Fifteen are empty for every labelled row and are dropped rather than imputed, since imputing a never-populated column fabricates a feature. Nineteen analytes remain: eight cannabinoid columns (`cbd`, `cbda`, `cbn`, `delta_8_thc`, `delta_9_thc`, `thca`, `total_cbd`, `total_thc`), ten terpenes (`alpha_bisabolol`, `alpha_humulene`, `alpha_pinene`, `beta_caryophyllene`, `beta_myrcene`, `beta_pinene`, `caryophyllene_oxide`, `d_limonene`, `linalool`, `terpinolene`) and `total_terpenes`.

Non-detects require care. The corpus has already normalised `ND` and `<LOQ` strings away, so only two cases remain:

- a numeric **zero** is a real measurement at or below the detection limit;
- an **empty** cell is an absence of information, either not tested or not reported.

These are not the same and must not be merged. The labelled dataset therefore keeps two columns per analyte, the value and a boolean `_tested` mask, and rows are never dropped for having gaps. The classifier receives the 19 value columns with empty cells left missing, which the estimator routes natively at each split, so the tested-versus-unttested distinction reaches the model through missingness rather than through the mask columns. The mask columns drive the coverage figures of Table S5 and the pre-check in 2.3.

The distinction produces two different coverage figures, computed in Table S5 on the rows that name a producer. Terpene panels are requested at similar rates for both classes, a gap of 0 to 5 points. The large gap, 19 to 27 points, is in *detection*: hydrocarbon extracts report a non-zero terpene value far more often. That gap is in what is reported once a panel is run, not in which test was ordered, and with 51.5% of the named non-solvent rows being distillates, refining is a likelier cause than the extraction category itself. The panel is not constant over time either. Nevada listed terpene analysis among the required tests for both categories in the text effective November 2020 and no longer does in the July 2022 text [11], and among the rows that name a producer the share with a terpene panel falls from 96.2% in 2020 and 96.5% in 2021 to 85.8% in 2022 and 72.0% in 2023.

One column couples the features to the grouping. `total_terpenes` is populated for a row exactly when the row names a producer, a laboratory convention and not a per-sample decision, so a blank value marks, inside the model, the rows that the unseen-producer scheme must drop. That coupling is why the headline comparison in 3.2 is made on the rows that name a producer. Section S4 documents it, together with the collinearity of `total_terpenes` with the ten terpenes and the unit artefacts in `total_thc`.

2.3 The leakage pre-check

A detection gap of 19 to 27 points is a candidate shortcut. If the pattern of which terpenes were reported, or reported as non-zero, tracked the class, a classifier could score well while reading the reporting convention rather than the chemistry. We measured this before modelling rather than assuming it away.

A classifier given **only the missingness pattern** of the 19 analyte columns the model sees, with every measured value discarded, reaches 69.9% accuracy against a 65.8% majority-class baseline: a lift of **+4.1 points**. Adding laboratory identity as an explicit feature reaches 70.5%, a further 0.6 points. Those are in-sample figures for the lookup. Cross-validated, the probe scores 69.8% plain and 56.8% balanced accuracy, against 84.1% plain and 81.0% balanced for the main model under a random split, so on the rows as stored the shortcut is 4.0 of 18.3 points on one metric and 6.8 of 31.0 on the other.

That lift is concentrated. Seven of the 13 missingness patterns the 19 columns take contain a single class, and they cover 11.04% of the usable rows, nearly all from the four laboratories that report a restricted analyte panel and record no producer. The shortcut is therefore a laboratory reporting convention and not chemistry, and much of it is an artefact of the split records of 2.1 (Section S5, Table S6). Those producer-less rows are also the rows the unseen-producer scheme must drop, which is why the headline comparison in 3.2 is made without them.

The cheapest shortcut therefore buys about 4 points, concentrated in 11% of the rows and attributable to reporting convention. That is small enough that a model scoring in the eighties cannot be running on it alone, which is what made the main experiment worth running, but it is not nothing, and 3.2 reports what happens when those rows are excluded.

2.4 Model and split schemes

A single model produces every headline figure: `HistGradientBoostingClassifier` (scikit-learn 1.9.1 [14]), `max_iter=200`, seed 20260921, early stopping disabled, fed the 19 analyte columns of 2.2. The point of the study is the evaluation protocol, not the estimator, so the estimator is held constant across all schemes and no hyperparameter is tuned; 3.2 varies its capacity instead. The alarm probes of 2.6 use the same estimator at `max_iter=120` and a learning rate of 0.05, for the reason given there. A standardised logistic regression (L2 penalty, $C = 1$) is run beside it in 3.2 and 3.3 as a low-capacity comparator. Since a linear model cannot route missing values, on the market data it receives $\log(1 + x)$ of each value with empty cells set to zero, plus one reported-or-not indicator per analyte. The headline model is unweighted, so 3.2 also runs both estimators with class weights inversely proportional to class frequency.

Four split schemes are compared on the market data:

1. **random**: stratified five-fold, the family of protocol that [1] uses;

2. **unseen-producer**: GroupKFold grouped by producer string, so no producer string appears in both train and test;
3. **unseen-lab**: leave-one-lab-out. Each fold holds out one entire laboratory. Only laboratories with at least 500 rows are eligible to be held out, because a fold testing on a 100-row laboratory estimates nothing; 8 of the 12 laboratories clear that floor and all 8 contain both classes, giving 8 folds. The 4 excluded laboratories hold 561 rows, 1.50% of the data, and they are still present in every training set;
4. **dual-class + unseen**: restricted to the 53 producers that make *both* classes, grouped by producer.

Scheme 4 is the strictest available test. Restricting to producers who make both classes removes the degenerate case where a producer's identity determines its label outright, but it does not make producer identity uninformative: the solventless share across these 53 producers ranges from below 0.01 to 0.99 with a median of 0.35, and 30 of the 53 sit outside the 0.2 to 0.8 band. A classifier given *only* an encoded producer ID, and no chemistry at all, still reaches 84.4% balanced accuracy inside this subset under a random split. What the restriction guarantees is that the held-out producer's label cannot be inferred from having seen that same producer in training, since every producer in the subset contributes both classes. It covers 27,751 rows, 74.3% of the usable data, with a 17,312 / 10,439 class split.

Three further analyses in 3.2 run on the rows that name a producer. Twenty reassignments repeat scheme 2 with the producers shuffled into folds, `GroupKFold(5, shuffle=True)` with seeds 0 to 19. A temporal split sorts the dated producer rows by `date_tested`, fits once on the earliest 80% and tests on the rest, without grouping; rows dated on the boundary day can fall on either side. A positive control replaces the target with whether the product name says distillate and runs schemes 1 and 2 unchanged.

On the controlled data, random five-fold is compared against leave-one-variety-out.

2.5 Metrics

The classes are roughly 1:2, so the extraction task is scored by balanced accuracy; where plain accuracy appears, in the 2.3 probe and the alarm probes of 2.6, it is given beside its majority-class baseline. Balanced accuracy has a chance level of 50.0% for two classes and 33.3% for three. Because it is computed at the default decision threshold, which an unweighted model on unbalanced classes does not place well, 3.2 also reports ROC AUC, which needs no threshold and has a chance level of 0.5. We also report macro F1, per-class recall, a confusion matrix, and the spread across folds. Every "+/-" in a table is the population standard deviation of the fold scores (divisor n). Producer-level intervals are percentile intervals over 2,000 bootstrap draws of producers, computed from the held-out predictions of the reported partition without refitting. A draw that contains one class only is skipped, and for a fold-mean statistic a fold that loses a class in a draw is left out of that draw's mean. We report eighteen such intervals, sixteen pooled (Table 5) and two on the fold mean (3.2, Figure 1b), without a multiplicity correction and call a bound within two points of chance marginal. Two baselines are computed explicitly: most-frequent (balanced accuracy 50.0%) and stratified random (50.7% on the one draw made, 50.0% in expectation). The alarm probes of 2.6 are multi-class and report plain accuracy against the explicit majority-class baseline of their target, with balanced accuracy beside it.

Per-class recall matters here because a single aggregate figure hides which side of the problem the model actually solves.

2.6 Alarm models

Before the main task, two probes asked whether the chemistry predicts metadata it should not: the testing laboratory and the producer. A profile that identifies the laboratory is a profile that partially encodes instrumentation, calibration and reporting convention, which would compromise any score obtained while train and test share laboratories. A third target, cultivar, has no reliable field in this corpus and could only be probed on a key reconstructed from product names; the limits of that probe are reported in 4.3 and the probe itself in Section S8.

Each probe keeps only classes with enough rows to be estimable, since a class with 20 members contributes noise rather than signal: at least 100 rows for the laboratory probe and 150 for the producer probe, the latter higher because producers are far more numerous. These floors, applied after the 30 conflicting rows are dropped, are why the probes in 3.1 run on 11 of 12 laboratories (37,276 rows) and 33 of 96 producers (30,416 rows) rather than on the full dataset. The probes receive the same 19 value columns as the main model, with empty cells left missing, so a laboratory's analyte panel is visible to them alongside its measured values. The laboratory probe therefore measures instrument and reporting convention together, which is the quantity that matters for a split that shares laboratories; 2.3 measures the reporting convention on its own, and 3.1 reports a missingness-only laboratory probe beside the chemical one.

The probes have 11 and 33 classes, and the softmax boosting diverges on them at scikit-learn's default learning rate of 0.1, so they are fitted at 0.05, where their fold scores are stable. The binary extraction task is far less sensitive to the rate, its held-out score moving by 0.1 points, and keeps the default (Section S6).

3. Results

3.1 The profiles identify the laboratory and the producer

Table 2. Alarm probes: predicting metadata from the 19 analyte columns, random five-fold.

target	classes	rows	accuracy	baseline	lift	bal_acc (chance)
lab	11	37,276	92.3% (+/-0.2)	40.3%	+52.0 pts	76.3% (9.1%)
producer	33	30,416	67.1% (+/-0.6)	14.4%	+52.7 pts	53.4% (3.0%)

The chemical profile identifies the testing laboratory on 92.3% of rows (Table 2), 52.0 points above the majority baseline, and the producer on 67.1%, 52.7 points above its own. Part of the laboratory signal is the reporting panel: a lookup table on the 13 missingness patterns alone, with no measured value, reaches 57.3% on the same folds, so the measured values add a further 35 points. The fingerprint is not producer memorisation either: grouped by producer on the seven laboratories with at least 100 rows that record one (33,161 rows), the laboratory probe scores 85.4% (+/-3.3) against a 45.3% baseline. This is expected, since instruments, calibration, reporting conventions and clientele all differ between laboratories, and this probe cannot separate them. The consequence is that any score obtained while train and test share laboratories is partly reading a laboratory-specific signal. With the split pairs of 2.1 merged the laboratory probe is unchanged at 92.3%, on ten laboratories and a 41.9% baseline.

3.2 Market data: 78.5% becomes 54.0%

Table 3. The reported model under each split scheme. random (optimistic) and the two baselines run on all usable rows; the row below it repeats the random split on the rows that name a producer.

split scheme	balanced_acc	macro_F1	folds
most-frequent baseline	50.0%	--	--
stratified baseline	50.7%	--	--
random (optimistic)	81.0% (+/-0.4)	81.8%	5
random, producer rows	78.5% (+/-0.4)	--	5
unseen-producer	54.0% (+/-5.6)	50.3%	5
unseen-lab	68.3% (+/-18.6)	67.6%	8
dual-class + unseen	54.6% (+/-2.6)	50.7%	5

The random (optimistic) row of Table 3 runs on all 37,344 usable rows and the two producer schemes only on rows that name a producer, so 81.0% and 54.0% are not like with like. On the same 33,229 rows the random split scores 78.5%, and that pair, 78.5% against 54.0%, a gap of **24.4 points** on the unrounded means, is the headline of this paper. The 81.0% includes the 4,115 producer-less rows, which carry the reporting-panel shortcut of 2.3 and the 1,303 samples stored twice (2.1); with those pairs merged it is 80.4%.

How far above chance the held-out score sits depends on how it is measured. The reported model is unweighted and the classes are roughly 1:2, so held out it answers mostly with the majority class, and balanced accuracy at the default threshold understates what its scores contain. Table 4 adds ROC AUC and class weighting, and Table 5 gives the pooled scores with their producer bootstrap intervals.

Table 4. Four models on the rows that name a producer: fold means under the random split and with the producer held out.

producer rows (33,229) model	random split		producer held out	
	bal_acc	AUC	bal_acc	AUC
boosting, unweighted	78.5%	0.894	54.0% (+/-5.6)	0.625
boosting, class-weighted	80.6%	0.894	55.8% (+/-5.5)	0.621
logistic, unweighted	65.3%	0.753	54.3% (+/-6.2)	0.634
logistic, class-weighted	69.4%	0.752	61.1% (+/-5.1)	0.636

Table 5. Producer held out: scores pooled over the held-out predictions of the reported partition, with producer bootstrap 95% intervals.

pooled, held out	bal_acc	AUC
producer rows (33,229)		
boosting, unweighted	51.6% [47.3, 56.6]	0.610 [0.535, 0.698]
boosting, class-weighted	56.8% [50.3, 63.0]	0.630 [0.551, 0.715]
logistic, unweighted	49.0% [43.0, 57.4]	0.604 [0.519, 0.696]
logistic, class-weighted	64.9% [56.8, 71.9]	0.662 [0.571, 0.745]
dual-class rows (27,751)		
boosting, unweighted	52.0% [45.8, 59.8]	0.594 [0.497, 0.704]
boosting, class-weighted	56.1% [48.2, 63.8]	0.615 [0.509, 0.723]
logistic, unweighted	51.7% [44.3, 60.7]	0.583 [0.480, 0.710]
logistic, class-weighted	61.2% [48.7, 71.6]	0.646 [0.519, 0.759]

Three things follow. First, a ranking signal transfers across producers: every model's held-out AUC is 0.62 to 0.64 as a fold mean, and on the producer rows the producer bootstrap interval of the pooled AUC excludes 0.5 for all four, marginally for the unweighted logistic model. Second, it is modest and does not always survive thresholding. The boosted models keep about 30% of the above-chance AUC the random split shows, 0.125 of 0.394 on fold means and 0.110 of 0.394 pooled for the reported one, the unweighted logistic model 41 to 53%, 0.134 and 0.104 of 0.253, and the class-weighted one 54 to 64%, 0.136 and 0.162 of 0.252, both from a much lower start. The pooled balanced accuracy of both unweighted models has an interval that includes 50%, and the unweighted logistic model's point estimate, 49.0%, is below it; only the class-weighted models clear chance, the boosted one marginally. For the reported model the fold-mean balanced accuracy of 54.0% has a producer bootstrap interval of [50.2, 58.9]. Third, the size of the loss depends on the estimator, on its capacity (Figure 1a) and on class weighting: 24.4 points for unweighted boosting, 24.8 with class weights, 11.0 for an unweighted logistic regression and 8.3 for a class-weighted one, whose held-out 61.1% is the highest fold-mean score of the four models.

The reported partition is one draw, and refitting on others moves the score. Twenty random reassignments of producers to folds (Figure 1b) give the reported model means from 51.3% to 56.6% (median 53.8%, sample sd 1.7; AUC 0.603 to 0.658) and the class-weighted one 53.0% to 59.9% (median 55.5%, AUC 0.602 to 0.664); none falls to 50.0%. The five folds of the reported partition score 48.6, 57.6, 46.7, 55.7 and 61.5. Laboratory identity alone, as a lookup under the same folds, scores 43.6%, below 50 because a laboratory's majority class among the training producers is a poor guide to the held-out ones, so the points above chance are not a laboratory prior.

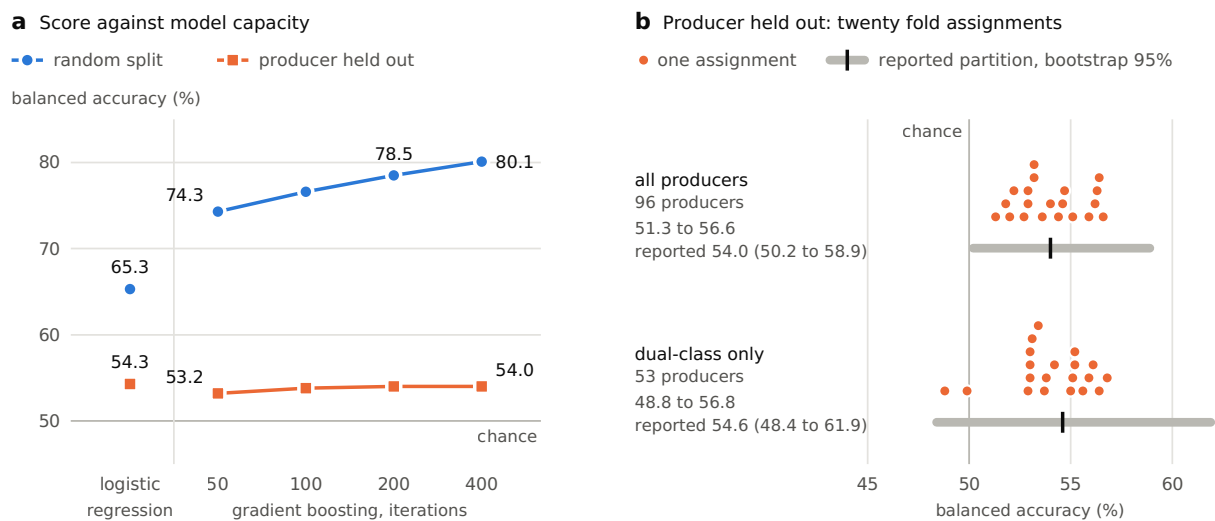


Figure 1. Balanced accuracy on the rows that name a producer, unweighted models. (a) Random split and producer-held-out score, for a standardised logistic regression and for gradient boosting at four iteration counts; fold means; the held-out fold spreads of the reported and logistic models are in Table 4. (b) Mean score of each of twenty random assignments of producers to folds, for all producers and for the dual-class subset; the bar is the producer bootstrap 95% interval of the fold-mean score on the reported partition and the tick is that score. Every plotted value is parsed from the evidence logs by `src/evaluation/make_figure.py`.

Table 6. Robustness of the gap. Balanced accuracy of the reported model; gap in points on unrounded means. The last row replaces the target (2.4).

check (producer rows, reported model)	random	held out	gap
reported: 200 boosting iterations	78.5%	54.0%	24.4
random-split seeds 1, 7, 12345			24.6, 24.5, 24.6
50 iterations	74.3%	53.2%	21.1
100 iterations	76.6%	53.8%	22.7
400 iterations	80.1%	54.0%	26.1
419 duplicate rows dropped	78.9%	55.2%	23.7
1,071 terpene-only rows dropped	79.4%	55.1%	24.3
two producer spellings merged		54.7%	
tested 2020-2022 only (23,461)	81.1%	52.6%	28.4
tested 2023 only (6,189)	83.5%	53.0%	30.5
rows not named distillate (19,166)	84.1%	60.6%	23.5
rows named for a process (4,345)	77.0%	60.3%	16.8
positive control: named distillate or not	89.6%	76.4%	13.2

Table 6 varies what could be suspected of producing the gap. It is stable under reseeding of the random split. It grows with model capacity while the held-out score does not move (Figure 1a), so 24.4 is one point on a curve from 21.1 to 26.1. Duplicate rows, rows that report only a zero terpene total and the two spellings of one producer do not produce it. Nor does the drift in the label and the panel (2.1, 2.2): the gap is 28.4 points on the rows tested in 2020 to 2022 and 30.5 on those tested in 2023. Two narrower definitions of the label keep more when held out, near 60% on a single partition each with fold spreads of 8.7 and 6.2: the rows not named distillate, and the rows whose names hit the mechanical family in the non-solvent class or the hydrocarbon family in the solvent-based class (leaving out the 19 solvent-based rows that also hit a mechanical word), the closest this corpus comes to rosin against BHO. The positive control shows that the hold-out does not destroy a score by itself: whether a row is named distillate, a target with a known chemical signature, is recovered at 76.4% for unseen producers.

Time has a cost of its own. Training on the earliest 80% of the dated producer rows and testing on the latest 20%, with 95.3% of the test rows from producers seen in training, scores 58.3% with an AUC of 0.754, against 79.5% and 0.902 for a random five-fold split of the same 29,650 rows. Balanced accuracy falls almost as far as under the producer hold-out and the AUC about half as far. Inside 2020 to 2022 the same split scores 64.6% with an AUC of 0.709. The two hold-outs are not interchangeable, and a model deployed on new producers at a later date would face both.

The strictest test agrees on the size of the loss. The dual-class subset scores 54.6% (+/-2.6) held out against 80.3% (+/-0.7) for a random split on the same 27,751 rows, a gap of 25.6 points on the unrounded means, and its AUC falls from 0.899 to 0.632. Its five folds on the reported partition are 54.1, 58.5, 53.2, 56.4 and 50.9, and the producer bootstrap of their mean spans [48.4, 61.9]; twenty reassignments give means from 48.8% to 56.8% (median 54.0%), two of them at or below 50.0%. On this subset no model's pooled balanced accuracy has an interval that excludes 50%, and only the class-weighted models' pooled AUC intervals exclude 0.5, marginally (Table 5). Laboratory identity alone scores 40.8% under the same folds. Macro F1 is 50.3% (+/-4.3) and 50.7% (+/-3.1) on the two schemes, against 39.8% and 38.4% for the majority predictor and 49.6% and 49.4% for a stratified shuffle on each scheme's own rows, so on that thresholded metric neither beats a stratified guess.

Held out, the reported model mostly answers with the majority class: pooled over the held-out producers its solventless recall is 29.6%, and its residual skill lives on the easy side (Section S7, Table S7).

Holding out the laboratory. The `unseen-lab` scheme has the highest mean of the three group-held-out schemes, 68.3%, and a fold-to-fold standard deviation of **18.6 points**: the score depends on which laboratory is held out, because holding out a laboratory shifts the test-set class prior as well as the instrument, and the two effects cannot be separated with these data (Section S9). We report it for completeness and do not rest any conclusion on it.

3.3 Controlled experiment: the same direction, an estimator-dependent size

Table 7. Controlled experiment, gradient boosting.

split scheme	balanced_acc	macro_F1	folds
chance level (3 classes)	33.3%	--	--
random 5-fold	56.7% (+/-9.2)	55.7%	5
leave-one-variety-out	38.9% (+/-12.6)	35.9%	6

With gradient boosting (Table 7), a random split suggests method is somewhat separable at 56.7%, and holding out an entire variety gives **38.9%**, 5.6 points above chance with a fold spread of 12.6 that straddles it. That is the market-data pattern with the held-out group changed from producer to variety. The mechanism is not the same: the design is balanced and has no replicates, so nothing leaks across a random split here, and what the variety hold-out measures is a shift between varieties. Nor does the drop survive a change of estimator intact:

Table 8. Controlled experiment, logistic regression.

logistic regression	balanced_acc
random 5-fold	54.3% (+/-2.5)
leave-one-variety-out	46.3% (+/-10.2)

A standardised logistic regression on the same folds (Table 8) drops 8.0 points instead of 17.8, which is inside its fold spread of 10.2, to a score that is above chance: 1 of 1,000 label shuffles reaches 46.3% ($p = 0.002$). Four features and about 130 training samples are a setting where a boosted ensemble can overfit each variety and a linear model cannot, so part of the boosted collapse belongs to the estimator. What both estimators agree on is the direction, and that the method signal which transfers across varieties is modest.

The variety is easier to read than the method:

chemistry -> variety:	balanced_acc 60.7% (+/-8.3)	chance 16.7%
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Under the same random split the four cannabinoids predict the **variety** at 3.6 times chance with gradient boosting (60.7%) and 2.8 times with logistic regression (46.9%), against 1.7 and 1.6 times chance for the method. Even in an experiment built to isolate method, with clean HPLC numbers, the variety, which here also means the batch of starting material, is read more easily than the method.

Three checks qualify the boosted figures, each recomputed from the raw file. First, the random-split score depends on the seed: over 200 seeds, five-fold scores average 54.3% (sd 2.7) and the reported 56.7% sits at the 80th percentile, so the seed-robust gap to leave-one-variety-out is 15.4 points rather than 17.8; for

logistic regression the 200-seed average is 51.8% (sd 2.3) and the gap 5.5. Second, the drop comes from the variety structure and not from the fold count: holding out six groups of 27 samples assigned at random, which keeps the fold geometry and breaks the variety link, averages 54.0% over 200 draws and never falls to 38.9%. Third, a permutation test over 300 global label shuffles, which also break the balanced design, gives $p = 0.0033$ for the random split, the floor for 300 shuffles, and $p = 0.11$ for leave-one-variety-out, so the boosted grouped score is not distinguishable from chance.

This matters for interpretation. The Cannlytics result alone could be dismissed as a consequence of messy market data. The controlled experiment shows the same direction on clean data, and it also shows that the size of a collapse is a property of the estimator as well as of the data.

4. Discussion

4.1 What this establishes

Two claims, of different kinds.

The **chemical claim** is narrow: the analyte panels of this corpus, 19 columns of which three are derived totals, heavily left-censored, carry only a weak signal of the declared extraction category, non-solvent against solvent-based (2.1), for a producer the model has never seen: an AUC of 0.62 against 0.89 under a random split, and 54 to 61% balanced accuracy depending on the estimator and on class weighting. Producer variation dominates whatever method signal exists: on the dual-class rows under a random split, a classifier given only an encoded producer ID reaches 84.4% and the chemistry 80.3%, while the chemistry grouped by producer reaches 54.6%. Cultivar plausibly does the same, but the two datasets disagree on the evidence and we do not claim it for the market corpus: in the controlled experiment the four cannabinoids read variety at 60.7% against 16.7% chance while reading method at 38.9% against 33.3% (46.9% and 46.3% with logistic regression), whereas in the market data the chemistry reads our reconstructed strain key only weakly, 9.9 points above its baseline (4.3, Section S8).

The **methodological claim** is general and, we think, the more useful one. Cannabis chemical data has group structure, and the validation protocols in use ignore it, a concern raised in general terms by [9] and given a size here. Two independent datasets, one market and one designed, both lose performance when the group is held out: on the market data 24 to 25 points of balanced accuracy with gradient boosting and 8 to 11 with logistic regression, and 0.27 of AUC for the reported model; on the designed experiment 15 to 18 points with gradient boosting and 5 to 8 with logistic regression. The mechanisms differ. In the market data the same producers sit on both sides of a random split; the designed experiment has no replicates, and its drop is a shift between varieties. The market gap is not produced by the 2023 change in labelling, since it is as large on the rows tested in 2020 to 2022 as overall; drift inside that window is not excluded. What the random split rewards is largely group identity, which the same profiles carry strongly: producer at 67.1% against a 14.4% baseline, laboratory at 92.3% against 40.3%, variety at 3.6 times chance.

4.2 What this does not establish

Not that extraction method leaves no chemical trace. Controlled studies with fixed genetics and fixed instrumentation measure method effects directly, and 3.3 reproduces one such effect under a random split; this work does not contradict them.

Not that a better model would fail. Only that these features generalise weakly across producers. A richer assay, a full terpene panel or an LC-MS fingerprint, is a different experiment.

Not that the labels are wrong, and not that they are clean. They are source-declared and agree with the name-derived path on 99.3% of the 11.5% of rows where both exist; that check covers eleven name patterns only, three for rosin and eight for hydrocarbon products. Product names cannot count mislabelled rows (2.1), and the treatment of ethanol extracts changed within the period (Table S4). What the labels mean is the regulator's category as applied (2.1), which is wider than the rosin-and-hash reading of "solventless": 3.2 gives the result on the rows that come closest to rosin against BHO, and a reader who wants that comparison specifically should rely on those and not on the headline.

Not chance, but close to it. On the producer rows the pooled AUC interval excludes 0.5 for all four models, and the pooled balanced-accuracy interval excludes 50% only for the class-weighted ones. On the dual-class subset no balanced-accuracy interval excludes chance, and only the class-weighted models' AUC intervals do (Table 5). Several of these bounds are marginal. The claim is about the size of the inflation, not that nothing transfers.

Not a joint hold-out. No scheme here holds out the producer and the laboratory together: the two largest laboratories hold 83% of the rows that carry a producer, so such a split cannot be built from this corpus.

Not a claim about the correctness of the conclusions in [1], [2] or [3]. Their chemical findings may well hold. What we show is that the performance figures of [1] and [2] are obtained under validation that leaves group structure intact, a family of protocols in which we measure substantial inflation for the random-split member, and so cannot be read as estimates of performance on a new producer, genotype or laboratory. We could not establish the protocol of [3] from its abstract.

4.3 The confounder we could not test cleanly

Strain is the obvious remaining candidate. Extraction routes are not applied to the same cultivars at the same rates, so the declared category may partly encode which cultivar was used, and cultivar drives terpene profile directly.

We could not test it cleanly. The `strain_name` and `strain_type` columns exist in the source file but are populated for **0 of 37,344** usable rows. The only cultivar signal left is in `product_name`, and a strain key extracted from it is too dirty to carry a confounder probe: 22.6% of the keys that parse are packaging or process words, not cultivar names (Table S8). Section S8 reports the three analyses we ran on that key, among them the alarm probe cited in 4.1, and we rest no conclusion on any of them. The strain question is open and cannot be answered with this corpus alone.

4.4 Recommendations

For anyone fitting a classifier to cannabis chemical data:

1. **Group by the entity production will hand you fresh.** Producer, genotype, laboratory. `GroupKFold` and `LeaveOneGroupOut` cost nothing to run.
2. **Report the random-split score alongside the grouped one.** The gap is the informative quantity and is cheap to obtain.

3. **Never report plain accuracy without its majority-class baseline.** Reference [1]'s 95.2% sits on a 78.8% baseline, so the lift is 16.4 points.
4. **Report per-class recall and the between-fold spread.** Our unseen-lab mean of 68.3% carries a standard deviation of 18.6; the mean alone is misleading.
5. **Run an alarm model before the real task.** If the chemistry predicts the laboratory, any laboratory-sharing score is partly laboratory-specific.
6. **Keep a tested-versus-detected distinction in censored panels.** Merging a measured zero with a missing cell changes the coverage gap in our data from 0 to 5 points to 19 to 27.
7. **Report the gap for more than one estimator.** Here it runs from 21 to 26 points across boosting iterations and is 8 to 11 for a linear model.
8. **Report a threshold-free metric and a class-weighted model.** Our unweighted model's held-out 54.0% hides an AUC of 0.62; class weighting alone moves a linear model from 54.3% to 61.1%.

4.5 Where a decisive answer would come from

The remaining untested avenue is a method-comparison dataset pairing extraction method with a full metabolomic fingerprint including terpenes, of the kind held in public LC-MS and GC-MS repositories such as MassIVE. That alone could distinguish "method leaves no usable trace" from "method leaves a trace, but not in the 4-cannabinoid or 19-analyte projections tested here".

5. Conclusion

The declared extraction category, non-solvent against solvent-based, is only weakly identifiable from the cannabinoid and terpene panels of this corpus of public laboratory results for an unseen producer. On the rows that name a producer, a random split reports 78.5% balanced accuracy and an AUC of 0.89; holding out the producer leaves 54.0% (51 to 57% across twenty fold assignments) and 0.62. Pooled over held-out producers the AUC is 0.61, with a producer-level bootstrap interval of 0.535 to 0.698 that excludes chance, while the reported model's pooled balanced accuracy, 51.6%, does not. A designed HPLC experiment shows the same direction when variety is held out, and the same profiles identify the producer and the laboratory far above their baselines.

The published accuracies cited here, 95% and 100%, are obtained under validation that holds out no group, the family of protocols for which we measure 24 to 25 points of inflation on market data and 15 to 18 on a designed experiment with gradient boosting, and 8 to 11 and 5 to 8 with a linear model. We ran random cross-validation, not Kennard-Stone selection, so the magnitude we report is specific to the former; what the two share is that neither holds out the producer, the genotype or the laboratory. We do not claim the published chemical conclusions are wrong, and we did not rerun those studies. We claim their numbers do not answer the question a reader will assume they answer. Group-aware validation is not a refinement here. It changes the size of the result by 8 to 25 points.

Data and code availability

Source data are public and are not redistributed here. The Cannlytics `cannabis_results` corpus [12] and Zenodo record 13823859 [13] are both CC-BY-4.0. The corpus is retrieved by `src/ingestion/fetch_cannlytics.py`, which verifies the downloaded size against the figure declared by the host API; the Zenodo file was downloaded by hand. The check matters: an early download truncated to 7% of its size, 177.9 MB of 2,534.8 MB, opened without error and was caught only by that check.

All analysis code and the raw evidence log for every run described in this manuscript are public at <https://github.com/ArtBreguez/cannabis-extraction-ds> (code MIT, manuscript CC-BY-4.0). The structure is:

<code>src/ingestion/</code>	retrieval with download-integrity verification
<code>src/preprocessing/</code>	labelling, unit checks, non-detect encoding
<code>src/models/</code>	the four split schemes and the alarm probes
<code>src/evaluation/</code>	recomputation and audit of every figure reported here
<code>docs/evidence/</code>	raw stdout of each run, committed unedited

A Supporting Information file accompanies this manuscript, with Sections S1 to S9 and Tables S1 to S8; its source is `SUPPORTING_INFORMATION.md` in the same repository and it is audited by the same script.

A single script, `src/evaluation/rederive_paper_numbers.py`, recomputes every dataset figure in this manuscript directly from the labelled dataset, and `src/evaluation/audit_preprint.py` checks each claimed number in this text against either that recomputation or the committed log, failing on any disagreement. The repository state that matches this manuscript is tagged `preprint-v1`.

Author contributions

A.G.B. is the sole author and is responsible for the conceptualisation, data curation, formal analysis, methodology, software, validation, visualisation and writing of this work. The assistance of AI tools is described below.

Funding

This work received no funding.

Competing interests

The author produces cannabis-related content on social media under the handle `@HiddenTerps` and has a commercial interest in mechanically separated (solventless) extracts, which fall inside the non-solvent category studied here. This work reports that the category is only weakly recoverable from routine laboratory panels, which runs against that interest rather than supporting it.

Use of AI tools

AI coding assistants were used throughout this work, under the author's direction and review. Their contribution was to the analysis code, the verification harness and the manuscript prose: writing and refactoring the scripts in `src/`, running the audit passes described above, reviewing the manuscript against the evidence logs, and drafting and revising text that the author then checked.

AI tools do not meet authorship criteria and are not listed as authors. The author is responsible for every claim, number and conclusion in this manuscript.

References

- [1] Freeman, T. P., Beeching, E., Craft, S., Di Forti, M., Frison, G., Lindholm, C., Oomen, P. E., Potter, D., Rieger, S., Rømer Thomsen, K., Zamengo, L., Cunningham, A., Groshkova, T., Sedefov, R. Applying machine learning to international drug monitoring: classifying cannabis resin collected in Europe using cannabinoid concentrations. *European Archives of Psychiatry and Clinical Neuroscience*, 275(2), 421-429, 2025. doi:10.1007/s00406-024-01816-w. PMC11910419.
- [2] Tran, J., Vassiliadis, S., Elkins, A. C., Cogan, N. O. I., Rochfort, S. J. Rapid In Situ Near-Infrared Assessment of Tetrahydrocannabinolic Acid in Cannabis Inflorescences before Harvest Using Machine Learning. *Sensors*, 24(16), 5081, 2024. doi:10.3390/s24165081.
- [3] Birenboim, M., Chalupowicz, D., Maurer, D., Barel, S., Chen, Y., Fallik, E., Paz-Kagan, T., Rapaport, T., Sadeh, A., Kengisbuch, D., Shimshoni, J. A. Multivariate classification of cannabis chemovars based on their terpene and cannabinoid profiles. *Phytochemistry*, 200, 113215, 2022. doi:10.1016/j.phytochem.2022.113215.
- [4] Kaufman, S., Rosset, S., Perlich, C., Stitelman, O. Leakage in data mining: formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), 15, 2012. doi:10.1145/2382577.2382579.
- [5] Kapoor, S., Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804, 2023. doi:10.1016/j.patter.2023.100804.
- [6] Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., Dormann, C. F. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913-929, 2017. doi:10.1111/ecog.02881.
- [7] Guignard, F., Ginsbourger, D., Levy Häner, L., Herrera, J. M. Some combinatorics of data leakage induced by clusters. *Stochastic Environmental Research and Risk Assessment*, 38(7), 2815-2828, 2024. doi:10.1007/s00477-024-02715-1.
- [8] Ambroise, C., McLachlan, G. J. Selection bias in gene extraction on the basis of microarray gene-expression data. *Proceedings of the National Academy of Sciences*, 99(10), 6562-6566, 2002. doi:10.1073/pnas.102102699.

- [9] Király, P., Tóth, G. Being Aware of Data Leakage and Cross-Validation Scaling in Chemometric Model Validation. *Journal of Chemometrics*, 39(4), e70026, 2025. doi:10.1002/cem.70026.
- [10] Kennard, R. W., Stone, L. A. Computer Aided Design of Experiments. *Technometrics*, 11(1), 137-148, 1969. doi:10.1080/00401706.1969.10490666.
- [11] Nevada Cannabis Compliance Board. Nevada Cannabis Compliance Regulations, Regulation 11, section 11.050, Required quality assurance tests; submission of wet cannabis for testing. Quoted from the version of 21 July 2022, https://ccb.nv.gov/wp-content/uploads/2022/07/Reg-11_v072122.pdf; the same wording stands in the version of 24 October 2023, https://ccb.nv.gov/wp-content/uploads/2023/11/Reg-11_v102423.pdf. The regulations effective November 2020, <https://ccb.nv.gov/wp-content/uploads/2021/02/Effective-NCCR-as-of-Nov-2020.pdf>, carry the same solvent-based definition and list terpene analysis for both categories. All accessed 2026-10-06. The 2018 predecessor, Nevada Administrative Code 453D.780 (regulation R092-17, effective 27 February 2018), named CO2 only.
- [12] Cannlytics. `cannabis_results`: public cannabis lab test results. Hugging Face Datasets, https://huggingface.co/datasets/cannlytics/cannabis_results, revision 8724df97f3c40b3f844157606dd91e97509a8cb9 (last modified 2026-02-02), file `data/all/all-results-latest.csv`, retrieved 2026-09-21. CC-BY-4.0.
- [13] Solís García, H. F., Suntaxi Crisanto, S. L. F., Vargas Delgado, L. F., De la Rosa Martínez, A. F., Londoño Larrea, P., González Benítez, D., Montúfar Delgado, C. Yields, Cannabinoids Quantification, and Predictive Programming Codes Using Machine Learning for Non-Psychoactive Cannabis Flowers and Extracts (*Cannabis sativa* L.) Cultivated in Ecuador. Zenodo, 2024. CC-BY-4.0. doi:10.5281/zenodo.13823859.
- [14] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830, 2011. <https://jmlr.org/papers/v12/pedregosa11a.html>